

Probability

Classical (theoretical) definition of probability, why it always lies between 0 and 1, elementary events, and the complement of an event, with worked examples and practice questions.

Introduction

Probability is a measure of how likely something is to happen. When you toss a coin, you are not sure whether you will get a head or a tail, but you know each is equally likely. When you roll a die, each face 1 to 6 is equally likely to show up. This chapter gives a precise way to write that likelihood as a number between 0 and 1, using the idea that all outcomes are equally likely.

This chapter deals with **theoretical (classical) probability**, the probability worked out by counting outcomes, assuming each outcome is equally likely. Experimental probability, which you saw in Class 9 by repeating an experiment many times, is not the focus here.

Basic Terms, Experiment, Event, Elementary Event

A **random experiment** is an action whose outcome cannot be predicted in advance, even though all possible outcomes are known. For example, tossing a coin, rolling a die, or drawing a card from a well-shuffled deck are random experiments.

Key terms

Outcome: a single possible result of a random experiment. For a die, the outcomes are 1, 2, 3, 4, 5, 6.

Elementary event: an event that has only one outcome. "Getting a 4 on a die" is an elementary event. Elementary events are the building blocks, every other event is made by combining them.

Event (E): a collection of one or more outcomes. "Getting an even number on a die" is the event {2, 4, 6}, which combines three elementary events.

Favourable outcomes: the outcomes that make the event happen. For "getting an even number", the favourable outcomes are 2, 4, 6.

Equally likely outcomes: outcomes that have the same chance of occurring. In a fair coin toss, head and tail are equally likely; in a well-shuffled deck, each of the 52 cards is equally likely to be drawn.

A die has six elementary events (1, 2, 3, 4, 5, 6). A coin has two (H, T). A deck of 52 cards has 52 elementary events.

Properties of elementary events

All elementary events of an experiment are equally likely and mutually exclusive (only one can happen at a time). The **sum of the probabilities of all elementary events is 1**.

For a die: $P(1) + P(2) + P(3) + P(4) + P(5) + P(6) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = 1$.

Classical (Theoretical) Definition of Probability

Classical definition

If all outcomes of an experiment are equally likely, the probability of an event E is:

$P(E) = \frac{\text{Number of favourable outcomes for E}}{\text{Total number of possible outcomes}}$

This is the only formula you need in this chapter. Every problem is just about counting favourable outcomes correctly and dividing by the total.

Worked Example 1, Die and coin

(a) A die is rolled once. Find the probability of getting an even number.

Total outcomes = 6 (1, 2, 3, 4, 5, 6). Favourable outcomes for "even" = 3 (2, 4, 6).

$$P(\text{even}) = \frac{3}{6} = \frac{1}{2}$$

(b) A coin is tossed twice. Find the probability of getting at least one head.

Total outcomes = 4 (HH, HT, TH, TT). Favourable for "at least one head" = 3 (HH, HT, TH).

$$P(\text{at least one head}) = \frac{3}{4}$$

Worked Example 2, Playing cards

One card is drawn from a well-shuffled deck of 52 cards. Find the probability of: (i) a king, (ii) a red card, (iii) a face card.

Total outcomes = 52. There are 4 kings, 26 red cards, and 12 face cards (4 jacks + 4 queens + 4 kings) in a deck.

$$P(\text{king}) = \frac{4}{52} = \frac{1}{13} \quad P(\text{red card}) = \frac{26}{52} = \frac{1}{2} \quad P(\text{face card}) = \frac{12}{52} = \frac{3}{13}$$

Remember: a deck has 4 suits, hearts and diamonds (red), clubs and spades (black). Each suit has 13 cards: Ace, 2, 10, Jack, Queen, King. Face cards are Jack, Queen, King.

Worked Example 3, Bag of balls

A bag contains 5 red balls, 3 blue balls and 2 green balls. One ball is drawn at random. Find the probability that it is (i) red, (ii) not blue, (iii) green or blue.

Total balls = 5 + 3 + 2 = 10. Each ball is equally likely to be drawn.

$$P(\text{red}) = \frac{5}{10} = \frac{1}{2}$$

$$P(\text{not blue}) = \frac{(10 - 3)}{10} = \frac{7}{10}, \text{ there are 7 balls that are not blue.}$$

$$P(\text{green or blue}) = \frac{(2 + 3)}{10} = \frac{5}{10} = \frac{1}{2}$$

Probability Is Always Between 0 and 1

For any event E, the number of favourable outcomes cannot be less than 0 and cannot be more than the total number of outcomes. So:

Range of probability

$$0 \leq P(E) \leq 1$$

$P(E) = 0$ means E is an **impossible event**, it can never happen.

$P(E) = 1$ means E is a **sure (certain) event**, it will definitely happen.

For a die rolled once:

- $P(\text{getting a 7}) = 0/6 = 0$, impossible, since the die only has faces 1 to 6.
- $P(\text{getting a number from 1 to 6}) = 6/6 = 1$, sure to happen.
- $P(\text{getting a 3}) = 1/6$, between 0 and 1, as expected.

A common mistake is to write probability as a percentage or ratio. In CBSE answers, always write it as a fraction or decimal between 0 and 1. So write $1/2$, not 50% or 1:2.

Also, do not confuse "equally likely" in the definition with always being $1/2$. The probability depends on how many favourable outcomes there are out of the total, and only when favourable = half of total does $P = 1/2$.

Complementary Events

For any event E, the event "**not E**" (written as $\bar{E}$ or E^c) is the event that E does not happen. E and "not E" are called complementary events, exactly one of them must happen.

Complement rule

$$P(E) + P(\text{not } E) = 1 \quad \text{i.e.} \quad P(\text{not } E) = 1 - P(E)$$

The sum of the probabilities of an event and its complement is always 1.

This is very useful. Sometimes it is easier to find $P(\text{not } E)$ than $P(E)$ directly.

Worked Example 4, Using the complement

A bag contains 15 balls, of which 6 are red. One ball is drawn at random. Find the probability that it is not red.

Method 1, Direct: Number of non-red balls = $15 - 6 = 9$. So $P(\text{not red}) = \frac{9}{15} = \frac{3}{5}$.

Method 2, Complement: $P(\text{red}) = \frac{6}{15} = \frac{2}{5}$. So $P(\text{not red}) = 1 - \frac{2}{5} = \frac{3}{5}$.

Both methods give the same answer. In questions like "at least one" or "not", the complement method is often shorter.

Another example: when a die is rolled, $P(\text{not getting a 6}) = 1 - \frac{1}{6} = \frac{5}{6}$, because getting 6 and not getting 6 are complements.

Note: if E is an impossible event ($P = 0$), its complement is a sure event ($P = 1$), and vice versa. The two together always cover all possible outcomes.

Summary

- **Random experiment:** an action with unknown outcome but known set of possible outcomes. **Elementary event:** an event with a single outcome.
- **Classical probability:** $P(E) = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$, when all outcomes are equally likely.
- **Sum of probabilities of all elementary events = 1.**
- **Range:** for any event E , $0 \leq P(E) \leq 1$. $P = 0$ is an impossible event, $P = 1$ is a sure event.
- **Complement:** $P(E) + P(\text{not } E) = 1$, so $P(\text{not } E) = 1 - P(E)$.
- Standard sets to remember: a die has 6 outcomes (1, 6), a coin has 2 (H, T), a deck has 52 cards (4 suits $\times$ 13 cards; 12 face cards; 26 red, 26 black).

Practice Questions

1. A die is thrown once. Find the probability of getting (i) a prime number, (ii) a number greater than 4, (iii) a number not greater than 4.
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2. Two coins are tossed together. Find the probability of getting (i) exactly one head, (ii) at most one head, (iii) no tail.

3. A bag contains 8 red, 5 white and 7 black balls. One ball is drawn at random. Find the probability that it is (i) white, (ii) not black, (iii) neither white nor black.

4. One card is drawn from a well-shuffled deck of 52 cards. Find the probability of (i) an ace, (ii) a black queen, (iii) not a face card.

5. The probability of winning a game is 0.6. What is the probability of losing it? What kind of event is "winning and losing taken together"?

6. A box contains 20 cards numbered 1 to 20. A card is drawn at random. Find the probability that the number on the card is (i) divisible by 3, (ii) a perfect square, (iii) not divisible by 3.